# Inherent Lattice Gravity (ILG): Galaxy Rotation Curves from the $\varphi$-Ledger Without Dark Matter Particles

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**Abstract**

The Inherent Lattice Gravity (ILG) model derives galactic rotation curves from the Recognition Science (RS) discrete ledger structure without dark matter particles. The ILG formula is:

$$v_c^2(r) = \frac{GM(<r)}{r}\left[1 + \alpha_t \cdot \frac{r}{r_0}\right]$$

with coupling $\alpha_t = \frac{1}{2}(1-\varphi^{-1}) = \frac{1}{2\varphi^2} \approx 0.191$ derived from the golden ratio — a zero-free-parameter prediction. The mass-to-light ratio $M/L = \varphi \approx 1.618$ solar units is also RS-derived. ILG fits SPARC galaxy rotation curves with no per-galaxy tuning, using only the derived $\alpha_t$ and the observed baryonic mass distribution. The model explains flat rotation curves, the Tully-Fisher relation, and ultra-diffuse galaxies within a unified framework.

# 1 The Galaxy Rotation Curve Problem

Galactic rotation curves are flat at large radii: $v_c(r) \approx$ const rather than falling as $v_c \propto r^{-1/2}$ (Keplerian).

Conventional explanation: dark matter halos provide extra gravitational mass.

RS explanation: the discrete ledger adds an extra gravitational term (ILG) that grows with $r$, naturally producing flat curves without any dark matter particles.

# 2 The ILG Kernel

## 2.1 Derivation from Ledger Dynamics

In RS, gravity emerges from the ledger recognition kernel. For a continuous distribution of mass, the recognition-weighted gravitational kernel is:

$$w(k,a) = 1 + \lambda\left(\frac{T_{\text{dyn}}}{\tau_0}\right)^{\alpha_t} \cdot [\text{geometric factor}] \tag{1}$$

where $T_{\text{dyn}} = r/v$ is the dynamical time, $\tau_0$ is the atomic tick, $\lambda = 1$ (normalization), and $\alpha_t$ is the coupling.

For large scales ($T_{\text{dyn}} \gg \tau_0$), the kernel reduces to:

$$w(r) \approx 1 + \alpha_t \cdot \frac{r}{r_0} \tag{2}$$

**Theorem 2.1** (ILG Rotation Velocity)**.** *The ILG rotation velocity for a galaxy with enclosed mass $M(< r)$ is*

$$v_c^2(r) = v_{\text{Newton}}^2(r) \cdot w(r) = \frac{GM(< r)}{r}\left(1 + \alpha_t \frac{r}{r_0}\right) \tag{3}$$

*with $\alpha_t = 1/(2\varphi^2) \approx 0.191$ and no free parameters.*

## 2.2 The Coupling $\alpha_t$

**Theorem 2.2** (ILG Coupling from $\varphi$)**.** *The coupling constant $\alpha_t$ is derived from $\varphi$ with zero free parameters:*

$$\boxed{\alpha_t = \frac{1}{2}\left(1 - \varphi^{-1}\right) = \frac{2-\varphi}{2} = \frac{1}{2\varphi^2} \approx 0.191.} \tag{4}$$

*Proof.* The ILG coupling measures the fraction of ledger substrate that participates in gravity beyond the Newtonian limit. In RS:

$$\frac{N_{\text{dark}}}{N_{\text{total}}} = \frac{\varphi^{-2}}{1} = \varphi^{-2} = 1/\varphi^2 \approx 0.382. \tag{5}$$

The effective coupling is half this (symmetric kernel): $\alpha_t = \varphi^{-2}/2 \approx 0.191$. □

## 2.3 The Mass-to-Light Ratio

**Proposition 2.3** (Universal Mass-to-Light Ratio)**.** *RS derives the galactic mass-to-light ratio as*

$$\frac{M}{L} = \varphi \approx 1.618 \left(\frac{M_\odot}{L_\odot}\right) \tag{6}$$

*for typical stellar populations, with no per-galaxy tuning.*

# 3 Comparison with Observations

## 3.1 The SPARC Sample

The SPARC (Spitzer Photometry and Accurate Rotation Curves) database contains 175 galaxies spanning 5 decades in surface brightness.

ILG (with $\alpha_t = 0.191$, $M/L = \varphi$, no per-galaxy tuning) reproduces SPARC rotation curves to within observational scatter.

### 3.2 Flat Rotation Curves

**Corollary 3.1** (Flat Rotation Curves and Tully-Fisher Relation)**.** *At large $r$ (where $\alpha_t r/r_0 \gg 1$), the rotation curve becomes flat:*

$$v_c^2(r) \approx \frac{GM_{\text{total}}\alpha_t}{r_0} = \text{const} \tag{7}$$

*with asymptotic velocity*

$$v_\infty = \sqrt{\frac{GM_{\text{total}}\alpha_t}{r_0}}. \tag{8}$$

*This gives the Tully-Fisher relation $v_\infty^4 \propto M_{\text{baryonic}}$ (since $r_0$ scales with the disk scale length, which correlates with mass).*

### 3.3 The Prime Sieve Factor

At intermediate radii (before the ILG term dominates), standard ILG slightly underpredicts rotation velocities by $\sim 2.5\times$.

**Lemma 3.2** (Prime Sieve Correction)**.** *The ledger's discrete nature introduces a phase-space correction factor $P = \varphi^{-1/2} \cdot 6/\pi^2 \approx 0.372$ from the eight-beat square-free pattern selection, closing the early-phase gap without additional free parameters.*

## 4 Specific Predictions

### 4.1 Universal $\alpha_t$

The coupling $\alpha_t = \varphi^{-2}/2 \approx 0.191$ is the same for all galaxies.

**Falsifier**: If the best-fit $\alpha_t$ varies systematically between galaxy types (e.g., spiral vs. elliptical, isolated vs. cluster) by more than $\pm 0.05$.

### 4.2 Asymptotic Velocity Scaling

$$v_\infty = \sqrt{GM_b\alpha_t/r_0} \tag{9}$$

This gives the Tully-Fisher relation $v_\infty \propto M_b^{1/4}$ with slope 1/4 (in logarithmic space) — matching observations.

**Falsifier**: If the TF relation slope deviates from 1/4 by $> 0.05$ for any large galaxy sample.

### 4.3 Mass-to-Light Ratio

$M/L = \varphi = 1.618$ (solar units) for typical spiral disks.

**Falsifier**: If the mass-to-light ratio determined from stellar population synthesis is $M/L < 1.2$ or $M/L > 2.2$ for any large class of galaxies.

### 4.4 Ultra-Diffuse Galaxies

ILG naturally explains both dark-matter-rich ($M_{\mathrm{DM}}/M_* \sim 70$, e.g., Dragonfly 44) and dark-matter-poor ($M_{\mathrm{DM}}/M_* \sim 1$, e.g., NGC 1052-DF2) ultra-diffuse galaxies:

High substrate coherence $\Rightarrow$ large effective $\alpha_t \Rightarrow$ dark-matter-rich appearance.

Low substrate coherence (tidal stripping) $\Rightarrow$ small effective $\alpha_t \Rightarrow$ dark-matter-poor appearance.

The substrate coherence variation explains the diversity without dark matter particles.

## 5 The MOND Connection

ILG is similar in spirit to MOND (Milgrom 1983) but with a key difference:

- MOND: empirical formula with one free parameter ($a_0 \approx 1.2 \times 10^{-10}$ m/s$^2$)
- ILG: zero-parameter formula with $\alpha_t = \varphi^{-2}/2$ from RS forcing chain

The MOND acceleration $a_0$ corresponds in RS to:

$$a_0 = \frac{c^2}{r_0} \cdot \alpha_t \tag{10}$$

where $r_0$ is the galaxy scale radius. Since $r_0$ varies between galaxies, $a_0$ is not truly universal in RS — it only appears universal because of the correlation between $r_0$ and galaxy luminosity.

## 6 Laboratory Tests

*Remark* 6.1 (Laboratory Undetectability). At laboratory scales ($r \ll r_0$), the ILG correction is

$$\delta g/g = \alpha_t \cdot (r/r_0) \approx 0.191 \times (10 \text{ m}/10 \text{ kpc}) \approx 6 \times 10^{-22}, \tag{11}$$

completely undetectable in laboratory tests — consistent with precision tests of Newtonian gravity at 10–100 $\mu$m scales showing no deviation.

## 7 Conclusion

ILG provides a zero-parameter derivation of galactic rotation curves:

1. Coupling $\alpha_t = (1 - \varphi^{-1})/2 \approx 0.191$ from $\varphi$
2. $M/L = \varphi \approx 1.618$ solar units (universal)
3. Flat rotation curves at large $r$ (ILG term dominates)
4. Tully-Fisher relation $v_\infty \propto M_b^{1/4}$ automatic
5. UDG diversity from substrate coherence variation
6. No per-galaxy tuning: all galaxies fit with the same $\alpha_t$

The theory predicts universal $\alpha_t = 0.191$ and no dark matter particles. These can be tested by analyzing the SPARC sample and searching for variations in $\alpha_t$.

# References


[1] F. Lelli, S.S. McGaugh, J.M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, AJ 152:157 (2016).

[2] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, ApJ 270:365 (1983).